

Product Carbon Footprint Study Report

Product Name: MDI Precursor

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: MDI Precursor

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - IPCC 2021

II. Quantitative Purpose

[Information provided by the user]

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg MDI Precursor.

2. System Boundary

MDI Precursor production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of MDI Precursor produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **3.597 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

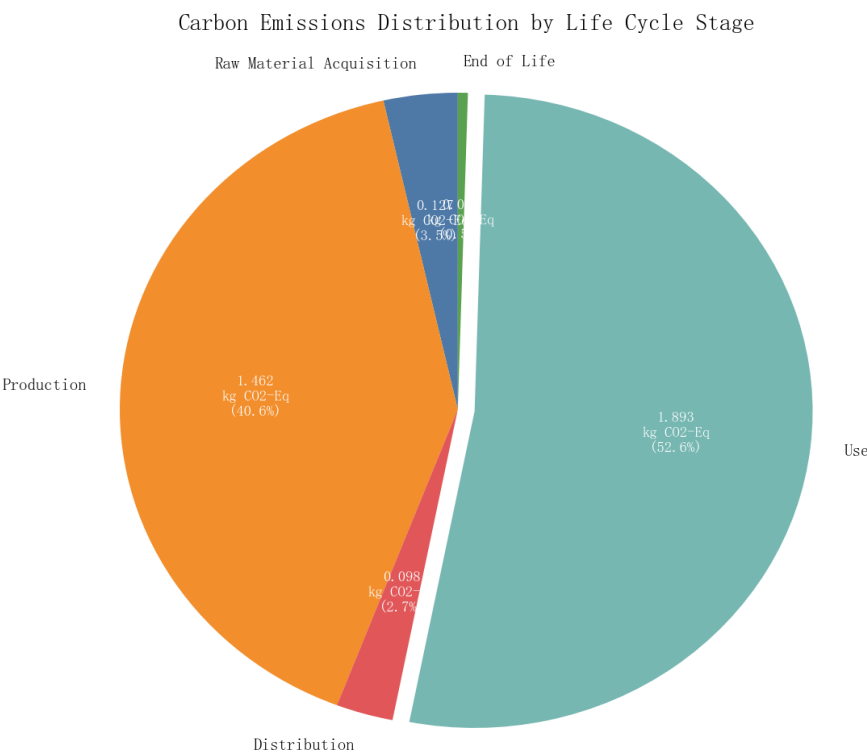


Table 1: Carbon Emissions by Life Cycle Stage

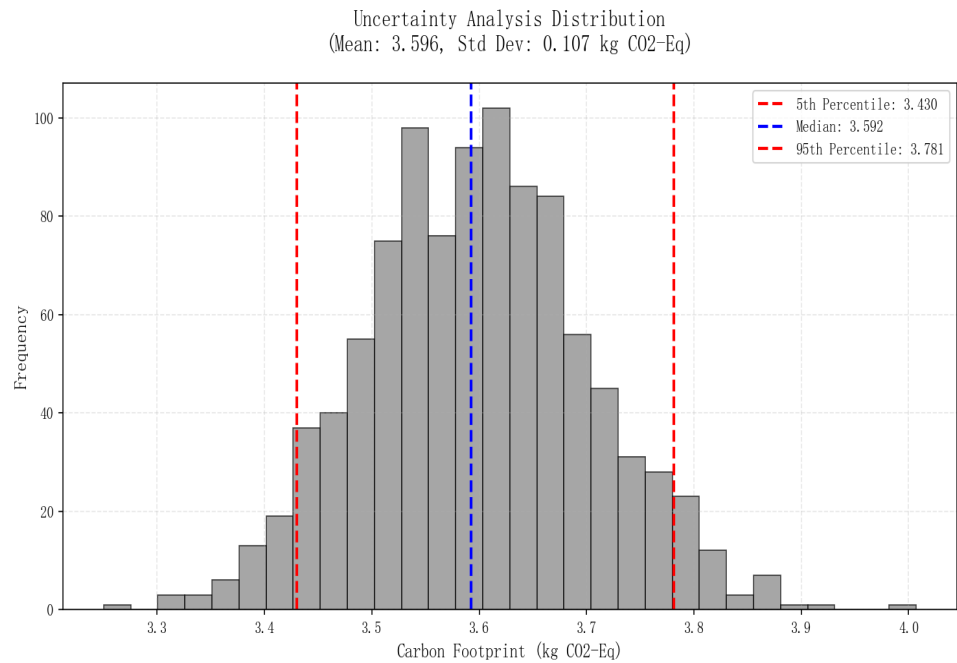
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.127	3.54
Production	1.462	40.64
Distribution	0.098	2.72

Use	1.893	52.62
End of Life	0.017	0.48
Total	3.597	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	3.596
Standard Deviation	0.107
Coefficient of Variation (CV)	2.97%
Median	3.592
5th Percentile	3.430
95th Percentile	3.781

VI. Interpretation of Results

1. Description of Results

The MDI Precursor exhibits a total carbon footprint of 3.597 kg CO₂-Eq, representing a moderate emission intensity typical for chemical precursors in industrial applications. This value reflects a cradle-to-grave assessment encompassing all major lifecycle phases, highlighting opportunities for targeted decarbonization within the chemical sector.

Lifecycle contributions reveal distinct patterns, with the use phase accounting for 52.6% (1.893 kg CO₂-Eq) and production for 40.6% (1.462 kg CO₂-Eq) of total emissions. Raw material extraction contributes 3.5%, distribution 2.7%, and end-of-life only 0.5%, indicating that downstream application and manufacturing processes dominate the environmental profile.

Major emission sources are concentrated in the use and production stages, which together comprise over 93% of the footprint. These phases likely involve energy-intensive reactions and operational energy consumption, underscoring the need for process optimization and energy efficiency improvements to achieve meaningful reductions.

2. Statement of Assumptions and Limitations

This study assumes representative average values for energy mixes, transportation distances, and operational parameters based on industry benchmarks rather than site-specific measurements. It further presumes standard product lifetimes and end-use scenarios consistent with typical MDI applications in polyurethane manufacturing.

Data sources primarily rely on secondary databases such as Ecoinvent, which may introduce variability due to regional differences or technological updates. Quality limitations include potential temporal mismatches and aggregation of proxy data for proprietary processes, reducing precision for certain inputs.

System boundaries are restricted to a cradle-to-grave scope covering raw material acquisition through disposal, excluding indirect effects such as capital equipment manufacturing, employee commuting, and land-use change impacts beyond the defined functional unit.

3. Suggestions for Improvement

- Implement energy efficiency upgrades and renewable energy integration in production facilities to reduce the 40.6% contribution from this stage.
- Optimize downstream use-phase applications through formulation adjustments or process innovations that lower energy demand during MDI conversion.
- Collaborate with suppliers to source lower-carbon raw materials, targeting the 3.5% raw material share with verified sustainable alternatives.
- Enhance end-of-life recycling infrastructure to increase material recovery rates and further minimize the minor 0.5% disposal emissions.
- Conduct primary data collection campaigns at production sites to improve inventory accuracy and enable more precise hotspot identification.

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